PRDA-01

STARTUP PROFIT ANALYSIS

Linear Regression & Power BI Dashboard

Prepared by	Manmeet Suthar
Project	Profit Analysis and Prediction
Tools	Orange Data Mining, Power BI, CSV data

Prepared from the model metrics, predictions, and dashboard details provided during the project workflow. Verify all values against the final source dataset before formal submission.

1. Executive Summary

This project explores how startup expenditure relates to profit and demonstrates a basic profit-prediction workflow. A Linear Regression model was evaluated in Orange Data Mining, two input cases were scored, and the outputs were presented in Power BI. The report includes a prediction-results page and a startup-profit-analysis page with state comparisons and scatter charts for R&D, administration, and marketing expenditure.

Reported evaluation metrics were R^2 = 0.856, MAE = 10,719.997, RMSE = 14,911.997, and MSE = 222,367,654.594. These results depend on the evaluation setup used in Orange and should be interpreted alongside its validation method and sample size.

2. Project Objectives

- Examine the relationship between startup spending and profit.
- Build and evaluate a Linear Regression model using Orange Data Mining.
- Generate profit predictions for two spending-input records.
- Visualize prediction outputs and startup spending/profit patterns in Power BI.
- Summarize findings and practical recommendations while noting analytical limitations.

3. Dataset Description

The project brief describes 50 startups in California, Florida, and New York. The stated fields are R&D Spend, Administration, Marketing Spend, State, and Profit. The spending columns are numerical inputs; State is categorical; Profit is the regression target.

Data availability note: The original dataset's row-level contents were not independently verified while generating this report. Confirm dataset completeness, units, missing values, and preprocessing against the actual file before submission.

4. Tools and Workflow

Stage	Tool / output	Purpose
Data access	Database / CSV export	Retrieve and prepare the project data.
Modeling	Orange Data Mining	Evaluate Linear Regression and generate predictions.
Visualization	Power BI Desktop	Build KPI cards, state comparison, and spending-vs-profit charts.
Deliverables	PBIX, OWS, CSV, report, screenshots	Package work and communicate results.

5. Modeling Method

Linear Regression estimates the relationship between numerical spending variables and Profit. The reported model used R&D Spend, Administration, and Marketing Spend as input features. The two prediction rows supplied three spending values each; no State values were provided for those rows, so the results reflect the model configuration actually used.

The supplied brief did not provide actual-profit labels for these two prediction cases. Therefore, their predicted values cannot be compared with actual profits using the available information.

6. Model Evaluation Results

Metric	Reported result	Interpretation
R^2	0.856	Explained about 85.6% of profit variation on the evaluated data.
MAE	10,719.997	Average absolute prediction error, in target units.
RMSE	14,911.997	Error metric that gives greater weight to larger errors.
MSE	222,367,654.594	Mean squared error, in squared target units.

Interpretation caution: R^2 is not prediction accuracy. Interpret these metrics with the validation method, small dataset size, and a baseline model in mind. Training-only evaluation can overstate performance.

7. Predicted Profit Results

Record	R&D Spend	Administration	Marketing Spend	Predicted Profit
Record 1	21,892.92	81,910.77	164,270.70	69,482.13
Record 2	23,940.93	96,489.63	137,001.10	70,825.71

Record 2 has a predicted profit 1,343.58 higher than Record 1. This is a model-based comparison, not proof that changing one spending category alone would cause that difference.

8. Power BI Dashboard and Observations

The report contains a prediction-results page and a startup-profit-analysis page. Visuals include prediction KPI cards, an input/prediction table, a State slicer, a profit-by-state bar chart, and scatter charts comparing spending with Profit.

- The R&D Spend vs Profit scatter chart appears to show a positive association in the displayed dataset.
- The Administration Spend vs Profit chart is more dispersed, so the relationship is less visually clear.
- The Marketing Spend vs Profit chart shows variation, with some exceptions to any overall pattern.
- The state bar chart compares total profit; average profit by state may be more suitable for comparing typical startup performance.

These are visual observations, not causal conclusions. Calculate correlations and state-level averages from the original dataset before presenting definitive findings.

9. Recommendations

- Validate the model using cross-validation or a held-out test set and compare it with a simple baseline.
- Check missing values, data types, outliers, and duplicate records before modeling.
- Use scatter charts and calculated correlations to quantify relationships between spending and profit.
- Compare both total and average profit by state; totals can be influenced by the number of startups per state.
- Do not treat predictions as guarantees or association as causation.
- Document data source, sharing permissions, and Orange workflow settings.

10. Limitations

- The project brief describes 50 startups, a small sample for broad conclusions.
- The validation setup for the reported model metrics was not supplied.
- The two prediction cases have no known actual-profit labels in the supplied brief.
- Currency/units and the original data source should be confirmed.
- Dashboard observations should be checked with numerical analysis.

11. Conclusion

This project demonstrates an analytics workflow combining Linear Regression in Orange Data Mining with reporting in Power BI. The model produced two profit estimates, and the dashboard enables comparison of those estimates and visual exploration of spending and profit. Before final submission, verify the original dataset, validation approach, and calculations; include screenshots of the final dashboard and saved Orange workflow.

12. Submission Checklist

- Power BI report saved as PRDA-01_Profit_Analysis.pbix.
- Orange workflow saved as PRDA-01_Linear_Regression.ows.
- Prediction output CSV included.
- Screenshots of both Power BI pages and Orange model evaluation included.
- Dataset included only if redistribution is permitted.
- No database credentials or passwords included.
- Report and presentation checked against final source files.

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End of report